MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

# Diagnosis and Management of Suspected and Definitive Necrotizing Enterocolitis

I. Applicability

a) Inclusion criteria: Intended for use in preterm infants with clinical concern for necrotizing enterocolitis (NEC) (see II below)

b) Exclusion criteria: Congenital GI anomalies, Term or near term infants

II. Clinical Presentation

Infants may present with a classical form of necrotizing enterocolitis (NEC) with abdominal distension (75%), bloody stool (28%) and bilious aspirates/emesis (18%).[1] Pneumatosis intestinalis and/or portal venous gas are pathognomonic radiographic findings.[2] NEC can present with a fulminant onset with shock and circulatory collapse, but can also present insidiously with non-specific signs (lethargy, temperature instability, or apnea). As infants with non-specific signs are more likely *not* to have NEC, this type of presentation offers a challenge to the clinician. For example, increased gastric residuals can be a hallmark of NEC, but can also be a normal finding in the premature infant. A significant overlap in the volume of residuals among infants with and without a subsequent diagnosis of NEC precludes it use as a marker for the disease.[3] [Please review the Brandon NICU feeding intolerance algorithm for more information]. Infants with other abdominal processes or sepsis may also present with these non-specific signs.

The possible cost of intervention (withholding of feedings, laboratory testing, x-rays, and antibiotics) must be weighed against the benefit of early treatment of those who will prove to have NEC. Using epidemiology to take into account the underlying a priori probability of NEC may help guide the clinician. Infants who are growth restricted (IUGR), whose mothers received multiple doses of prenatal steroids, who have underlying reasons for low cardiac output and ischemia, who received indomethacin, H2 blockers, and > 5 days of empirical antibiotic therapy may be at increased risk for NEC and non-specific signs taken more seriously.

III. Physical assessment of infants suspected of having NEC

a) Vital signs, cardiac/respiratory/neuro status

b) Hemodynamic status (capillary refill, blood pressure, urine output)

c) GI function and exam (bowel sounds, distension, color, girth)

IV. Interventions if NEC remains highly suspected:

a) Hold enteral feedings

b) Gastric decompression: Replogle to continuous suction

c) Radiographic evaluation: anterior-posterior and left lateral decubitus x-rays (cross table if unable to tolerate)

   i) Refer to “radiologic signs” in Table 1

d) Serologic tests

   i) If NEC is suspected but not definitive, consider sepsis evaluation (blood, urine, CSF), stool for guaiac, CBCPD, CRP, and basic metabolic panel

   ii) If definitive or advanced NEC based on x-ray and clinical presentation: blood gas, CBCPD, CRP, basic metabolic panel, blood culture

e) Volume resuscitation

i) Infants with changes in hemodynamic status or severe illness may need up to 2x volume replacement in severe disease[1]

ii) Since VLBW infants do not tolerate excessive fluids well, meticulous attention to intravascular water content should be made

f) Pediatric surgery consultation for infants with stage II or III NEC

   i) With serious and/or advancing disease, decisions regarding the need for operative intervention require surgical consultation

ii) To assist in making such decisions, earlier consultation (prior to the infant requiring critical care), will help the surgical team follow the progression of disease

g) For NEC requiring surgical intervention[5]
   i) Infants ≤1000 grams: primary drain placement
   ii) Infants >1500 grams: exploratory laparotomy
h) Pediatric infectious disease consultation:
   i) In cases of bacteremia or if no improvement/worsening within 48 hours of illness
   ii) Antibiotic choice is provided in Table 1 based on University of Michigan Health System antibiogram and in consultation with Pediatric Infectious Disease[5, 6]

V. Additional Considerations:
a) Laboratory testing frequency will be based on the trend and derangements present. For moderate to severe disease, derangements can be sudden and severe, especially with advanced disease.[1]
b) A systematic review of serological tests in the diagnosis of NEC reported: CRP was a "relatively sensitive but nonspecific marker for NEC", thrombocytopenia occurs in many neonatal systematic disease states, and nadir platelet counts typically occur after a diagnosis is made and hence are less useful in diagnosis.[7] CRP has been shown to be abnormal in infants with stage II and III NEC by 24-48 hours but may also be positive in other disease states (e.g., sepsis).[8]
c) Progression and/or perforation is most likely to occur within 48 hours of disease onset.[1, 9]
d) Abdominal ultrasound may be used as an adjunct study for infants with IIB or IIIA disease with inconclusive radiographs.[10]
e) Nasogastric suction may be discontinued when output is < 3 ml/kg/shift (opinion)
f) Contrast studies are not indicated routinely before re-starting enteral feeds and should be performed if evidence of feeding intolerance and the suspicion of stricture.[11]
g) Antibiotic levels must be monitored closely to minimize toxicities.[1, 12]

VI. Related Policies/Guidelines/Standards/Procedures
a) Feeding intolerance algorithm

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## Table 1: Staging and Management of Necrotizing Enterocolitis

| STAGE | ILLNESS SEVERITY | SYSTEMIC SIGNS | INTESTINAL SIGNS | RADIOLOGIC SIGNS | SURGERY CONSULT | X-RAYS FREQ. | ANTIBIOTICS[2,3] (see notes below) | DURATION |
|---|---|---|---|---|---|---|---|---|
| **I. Suspected – No radiographic evidence. Differential includes ileus, fissure, other systemic infections, etc.** | | | | | | | | |
| Ia & Ib | Suspicious, mildly ill | - Temperature instability<br>- Apnea/bradycardia<br>- feeding intolerance | Residuals, mild distension, occult blood (1a), gross blood (1b) | Normal or mild ileus | No | Q12 x 24-48 hours | **Ampicillin**, or **vancomycin** if central line or MRSA colonized and **gentamicin** | 48 hours |
| **II. Medical NEC – MUST HAVE RADIOGRAPHIC DIAGNOSIS** | | | | | | | | |
| IIa. | Mild medical NEC | - Same as Ia<br>- may (or may not) see some minor electrolyte or platelet changes | - Same as stg I<br>- +/- bowel sounds<br>- +/- abdominal tenderness | - Pneumatosis intestinalis (PI)<br>- or fixed dilated loops | Yes | Q12 x 48 hours | **Ampicillin**, or **vancomycin** if central line or MRSA colonized (change to ampicillin if cultures negative at 48hrs) and **gentamicin** | 7 days |
| IIb. | Moderate medical NEC | - Same as Ia, plus some lab changes (mild metabolic acidosis, mild thrombocytopenia)<br>- often needs more support (resp or CV) | - Abd tenderness<br>- Abd distension<br>- Absent bowel sounds<br>- +/- abdominal cellulitis | - Same as IIa<br>- often portal venous gas<br>- ± ascites | Yes | Q8 x 48 hours | **Vancomycin** (discontinue if cultures negative at 48hrs) and **piperacillin-tazobactam** | 10 days |
| IIIa | Severe medical NEC (very sick like IIIb) | - ***Mod-severe* metabolic and/or respiratory acidosis**<br>- electrolyte & CBC abnormalities<br>- **shock** | - As above<br>- + peritonitis,<br>- marked tenderness | - Same as IIb<br>- May see persistent ileus or absent bowel gas | Yes | Q8 x 48 hours | **Vancomycin** (discontinue if cultures negative at 48hrs) and **piperacillin-tazobactam** | 10 days |
| **III. Surgical NEC** | | | | | | | | |
| IIIb | Surgical NEC | - Same as IIIa | - Same as IIIa | Pneumoperitoneum | Yes | X-rays prn | **Vancomycin** (discontinue if cultures negative at 48hrs) and **piperacillin-tazobactam** and **fluconazole** | 14 days |

1. Systemic and intestinal signs are provided for your reference but generally reflect the illness severity (with moderate NEC, the systemic signs will be more deranged than with mild NEC). Not all signs will be present and crossover between stages may occur. Choose the stage that overall best fits the severity of illness.
2. If bacteremia present OR persistent symptoms/worsening at 48 hours, consult Pediatric Infectious Disease

Surgical intervention may be warranted if no clinical improvement after 48-72 hours and abdominal exam/X-rays remain concerning. Consider drain or paracentesis/ultrasound as a diagnostic study if NEC diagnosis unclear.

| **Last revised:** 02/2017 | **Antimicrobial Stewardship Subcommittee:** N/A | **P&T:** N/A |
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## References


| | Citation |
|---|---|
| 1 | Walsh M, Kliegman R. Necrotizing enterocolitis: treatment based on staging criteria. *Pediatr Clin North Am*. 1986;33(1):179-201. |
| 2 | Neu J, Walker WA. Necrotizing enterocolitis. *NEJM*. 2011;364(3):255-264. |
| 3 | Cobb B, Carlo W, Ambalavanan N. Gastric residuals and their relationship to necrotizing enterocolitis in very low birth weight infants. *Pediatrics (Evanston)*. 2004;113(1):50-53. |
| 4 | Necrotizing enterocolitis (NEC) guideline team. Evidence based care guideline: necrotizing enterocolitis (NEC) among very low birth weight infants. Pediatric evidence-based care guidelines: Cincinnati Children's Hospital Medical Center; 2010. p. 1-10. |
| 5 | Downard C, Renaud E, St Peter S, Abdullah F, Islam S, Siato J, et al. Treatment of necrotizing enterocolitis: an American Pediatric Surgical Association Outcomes and Clinical Trials Committee systematic review. *J Pediatr Surg*. 2012;47(11):2111-2122. |
| 6 | Shah D, Sinn J. Antibiotic regimens for the empirical treatment of newborn infants with necrotising enterocolitis. *Cochrane Database Syst Rev*. 2012(8):CD007448. |
| 7 | Evennett NN, Alexander N, Petrov M, Pierro A, Eaton S. A systematic review of serologic tests in the diagnosis of necrotizing enterocolitis. *J Pediatr Surg*. 2009;44(11):2192-2201. |
| 8 | Pourcyrous M, Korones S, Yang W, Boulden T, Bada H. C-reactive protein in the diagnosis, management, and prognosis of neonatal necrotizing enterocolitis. *Pediatrics (Evanston)*. 2005;116(5):1064-1069. |
| 9 | Najaf T, Vachharajani N, Warner B, Vachharajani A. Interval between clinical presentation of necrotizing enterocolitis and bowel perforation in neonates. *Pediatr Surg Int*. 2010;26(6):607-609. |
| 10 | Bohnhorst B, Kuebler J, Rau G, Gluer S, Ure B, Doerdelmann M. Portal venous gas detected by ultrasound differentiates surgical NEC from other acquired neonatal intestinal diseases. *Eur J Pediatr Surg*. 2011;21(1):12-17. |
| 11 | Wiland E, South A, Kraus S, Meinzen-Derr J. Utility of gastrointestinal fluoroscopic studies in detecting stricture after neonatal necrotizing enterocolitis. *J Pediatr Gastroenterol Nutr*. 2014;59(6):789-794. |
| 12 | Bhatt-Mehta V, Schumacher R, Faix R, Leady M, Brenner T. Lack of vancomycin-associated nephrotoxicity in newborn infants: a case-control study. *Pediatrics (Evanston)*. 1999;103(4):e48. |


## Revision History

| Date | Revision |
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*The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.*